

Benjamin Tran-Pugh

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EDUCATION

University of Waterloo

Candidate for BAsC, Mechanical Engineering (Co-op)

Cumulative GPA: 3.83/4.0

Sep 2025 – Jun 2030

Waterloo, ON

TECHNICAL SKILLS

CAD & Analysis: SolidWorks (FEA, PDM, Sheet Metal), Onshape (FEA), Fusion 360, Autodesk Inventor, AutoCAD, GD&T

Software: Python (NumPy, SciPy, Matplotlib), C/C++ (ESP32), MATLAB, Linux (Raspberry Pi, Jetson Orin Nano), Git

Other: 3D Printing (PLA, PETG, TPU, ABS, PA6-CF), DFM/DFA, Machining, Technical Drawings, Microsoft Suite, URDF

EXPERIENCE

Mechanical Engineering Intern

Human Computer Lab (SPC '25)

Feb 2026 – Aug 2026

Toronto, ON

- Owned mechanical and actuation development of LeLamp, an animatronic robotics product, across three actuator generations, advancing it from functional prototype toward manufacturing readiness through EVT/DVT stages
- Designed and hardware-validated a spring-based gravity-compensation system to eliminate actuator thermal overload, deriving joint balance conditions from Lagrangian mechanics and verifying against published literature
- Designed antagonistic-pair, capstan-drive, and full/partial gravity-compensation mechanisms, validating each by calculation, physics simulation, and iterative testing of hardware against predictions
- Led actuator trade studies with Pugh matrices and kill criteria; recommendations adopted into the product roadmap
- Identified reliability gaps in fatigue and lifecycle coverage and drove their integration into the validation roadmap from program start, de-risking late-stage failures, safety issues, and redesign
- Diagnosed root causes of motor noise from first principles and damped the dominant resonances in adjacent walls

Mechanical Team Member

Waterloo Aerial Robotics Group

Sep 2025 – Present

Waterloo, ON

- Designed and fabricated a sub-60 g payload-release mechanism for the Valkyrie competition drone, securing objects of varying size and mass in flight and releasing them at a small target with snag-free, propeller-safe routing; flew at AEAC 2026, where the team earned the Pip Rudkin Perseverance Award
- Designed iterations of empennage structures for Helios, a fixed-wing aircraft evolving toward VTOL capability
- Managed design data in SolidWorks PDM across a multi-hundred-member team, integrating mechanical designs with electrical and autonomy subsystems

PROJECTS

Double Pendulum Chaos Simulation

Python, SciPy, NumPy

- Derived the full nonlinear equations of motion by hand via the Euler–Lagrange equations, obtaining closed-form angular accelerations through Cramer's rule
- Built a numerical simulation (SciPy solve_ivp) demonstrating chaotic divergence from initial-condition perturbations as small as $1e-9$ rad
- Computed and visualized a time-to-flip fractal across the initial-condition space using vectorized NumPy

4-DOF Robotic Arm 2.0

SolidWorks, ESP32, C++

- Modeled a 50+ part assembly integrating off-the-shelf components; validated 8 structural components with FEA against load requirements, ensuring a factor of safety of 1.5
- Implemented 4-DOF inverse kinematics, trajectory smoothing, and a web-based control interface in C++ running on an ESP32, served over its built-in WiFi
- Wrote Python and MATLAB simulations to verify joint limits and workspace before hardware runs, preventing collisions, tipping, and overheating

Pneumatic Launcher

Fusion 360

- Designed a pneumatic launch system achieving 60+ m range on a \$100 budget; sized the pressure chamber to a factor of safety of 2 on burst pressure